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Original article

Efficacy of CBD-enriched medical cannabis for treatment of refractory epilepsy in children and adolescents – An observational, longitudinal study

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Abstract

The objective of this observational study was to evaluate the efficacy of medical cannabis for the treatment of refractory epilepsy. Fifty-seven patients (age 1–20 years) with epilepsy of various etiologies were treated with Cannabis oil extract (CBD/THC ratio of 20:1) for at least 3 months (Median follow up time-18 months). Forty-Six Patients were included in the efficacy analysis. Average CBD dose was11.4 mg/kg/d. Twenty-six patients (56%) had ≤50% reduction in mean monthly seizure frequency. There was no statistically significant difference in response rate among various epilepsy etiologies, and cannabis strain used. Younger age at treatment onset (<10 years) and higher CBD dose (>11 mg/kg/d) were associated with better response to treatment. Adverse reactions were reported in 46% of patients and were the main reason for treatment cessation.

Our results suggest that adding CBD-enriched cannabis extract to the treatment regimen of patients with refractory epilepsy may result in a significant reduction in seizure frequency according to parental reports. Randomized controlled trials are necessary to assess its true efficacy.
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Keywords: Seizures; Refractory epilepsy; Antiepileptic drugs; Anti-seizure drugs; Efficacy; Cannabis; Cannabidiol; CBD-enriched cannabis extract

1. Introduction

Twenty-five to 30% of the patients with epilepsy are pharmacologically resistant. Treatment resistance is defined as failure to control seizures after an adequate trial of at least two appropriate medications [1]. Second line therapeutic options are then available, albeit with partial efficacy in most cases, resulting in a substantial deleterious effect on individual health and quality of life and a heavy burden on society [2]. For this reason, there is a need for additional treatments options.

Marijuana (cannabis sativa and cannabis Indica) is a naturally growing plant containing more than 130 phytocannabinoids and 300 non-cannabinoid constituents [3,4]. Two compounds – the psychoactive Δ9-tetrahydro-cannabinol (THC), and the non-psychoactive cannabidiol (CBD), first described by Mechoulam and Shvo [5], have garnered the most attention based on their abundance in the plant. Both THC

Abbreviations: CBD, Cannabidiol; AEDs, Anti-epileptic drugs; ARs, Adverse reactions

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and CBD were found to have anticonvulsive properties in both in vitro and animal models [6–8], while most studies suggest that CBD has a superior role in reducing of epileptic activity compared to THC [9–11], which has a higher potential for cognitive impairment and chronic psychiatric disturbances [12]. The antiepileptic mechanism of CBD has not been fully elucidated and is considered to be mediated by inhibiting excitatory glutamatergic neurotransmission, mostly via cannabinoid receptor-independent mechanisms [7,10,11,13–20]. Other properties of CBD, including neuroprotective, anti-inflammatory and antioxidant properties [4] reviewed in 8, have been described. The effectiveness of marijuana in the treatment of epilepsy was originally reported as early as 1800b.c [21]. The first clinical trials using CBD in humans were described during the seventies [4,22]. It was in the 1990 s, when the discovery of an endogenous cannabinoid-signaling system rekindled the interest in the potential therapeutic effect of cannabis for treating nervous system disorders, including epilepsy. Since 2013, clinical studies have multiplied [23–28], and at present, there are about 25 ongoing clinical trials studying the effect on seizure frequency of CBD-enriched products, as well as their safety and drug interactions [http://clinicaltrials.gov]. Even though cannabis use is illegal in most countries, recent evidence suggests its potential efficacy in the treatment of epilepsy, particularly in the refractory group [23–28]. We report our findings following an observational, longitudinal study on cannabinoids for the treatment of refractory epilepsy in a single pediatric epilepsy clinic.

2. Material and methods

2.1. Subjects and study design

Children and adolescents (age 1–20 years) with refractory epilepsy, who were ambulatory treated at a tertiary pediatric medical center, were offered treatment with cannabis in addition to other accepted therapeutic options. Treatment-resistant epilepsy was defined as failure of at least 4 antiepileptic drugs, including ketogenic diet or Vagal nerve stimulation. All patients were receiving stable doses of antiepileptic drugs for at least 4 weeks before enrolment.

Exclusion criteria included previous or current treatment with cannabis-based therapy.

Patients who opted for cannabis treatment were prospectively followed by a single pediatric neurologist. Those with at least 3 months of follow-up were included in the efficacy analysis by intention-to-treat analysis. Seizure frequency was recorded for all patients based on parental or caregivers report before initiating cannabis treatment and on subsequent visits.

A calculation of pre-cannabidiol baseline mean monthly seizure frequency was performed based on the number of seizure per period of time (day/week/month) on the 8 weeks prior to treatment initiation, as reported by parents. Seizure frequency was calculated in each consecutive clinic visit.

CBD-enriched cannabis oil administered sublingually was supplied by two licensed growers (Better and Tikun Olam, Tel Aviv, Israel). Oil containing a CBD/THC ratio of 20:1 was extracted from one of two strains ("Cheesepie (EP1)" and "Avidekel") as described by Tzadok et al [25]. The preparation at the crude extract level, the purified CBD, and the final solution level were analyzed by high-performance liquid chromatography and gas chromatography-mass spectrometry. The ratio between CBD and THC was standardized and corrected to 20:1 by the addition of pure CBD. The CBD and THC analyses for each batch were performed in two independent labs approved by the Ministry of health (a university lab and a GMP-approved lab). The two strains of cannabis were analyzed several times during the study period. The ratio of THC: CBD remained unchanged throughout that period. There were minor variations in the concentrations of the CBD extract, which did not exceed 3%. Laboratory reports were provided to the investigators.

CBD-enriched cannabis oil at a starting dose of 2–5 mg/kg/day divided into three daily doses was added to the baseline antiepileptic regimen. Per protocol, dosage increments were performed until intolerance or maximum dose of 50 mg/kg per day of CBD. However, none of the patients reached the maximal dose of CBD. The concomitant AEDs were withdrawn per treatment response. THC dosage did not exceed 0.15–1.35 mg/kg/d.

Response to treatment was evaluated by comparing mean monthly seizure frequency at the last follow-up with pre-cannabis mean seizure frequency. The primary efficacy outcome was a reduction in mean monthly seizure frequency, for all countable seizure types, which included: primary generalized tonic-clonic seizures, atypical absence, complex partial seizures, myoclonic seizures, and partial seizures with secondary generalization. Seizure reduction was rated based on parentally reported number of seizures per period (0%, <49%, 50–75%, and 75–100%). A Positive response to treatment was defined as more than 50% decline in seizure frequency compared to baseline.

Additional data including improvement in spasticity in children with cerebral palsy, behavior changes, shortening of seizure duration or the duration of the post-ictal state were marked.

The study was approved by the IRB committee of the Tel Aviv Sourasky Medical Center (IRB# 0657–14).

Informed consent was obtained following a full explanation of the treatment provided.

2.2. Statistical analysis

Patients data was anonymized for purposes of statistical analysis.

For the sake of statistical analysis, patients were grouped into two broad groups-no response and <49% reduction in mean monthly seizure frequency, and 50–75% and 75–100% reduction in mean monthly seizure frequency. We used a two-sample *t*-test, Fisher's exact test or Chi-square test and binary logistic regression to analyze factors affecting response to treatment. Cox Proportional Hazard regression survival analysis was performed to evaluate the impact of variables on treatment duration, including response to treatment, and presence of adverse reactions. $P < 0.05$ was considered statistically significant. Statistical analysis was performed using the SPSS version 23 (SPSS Inc. Chicago, IL, USA).

3. Results

Between March 1, 2014, and June 30, 2016, sixty-nine patients initiated treatment with CBD-enriched cannabis oil. Twelve patients were excluded from the efficacy analysis: five due to inadequate parental reporting of seizures and seven patients who did not complete 3 months of follow up. Fifty-seven patients who were treated for more than three months or dropped out of the study due to ARs or treatment inefficiency prior to three months of treatment were included in the safety analysis. Eighteen of the patients were included in a previous multicenter publication, however with significantly shorter follow-up [25].

3.1. Study population (Table 1)

Patient's average age was 9.6 years (SD 4.9). Eighty percent of them were younger than fifteen years. All patients had refractory epilepsy due to various etiologies (Table 1). Patients were concomitantly taking up to 4 AEDs during the study period; Levetiracetam (17 patients), Clobazam (12 patients), Valproic acid (9 patients), Phenobarbital (9 patients), Topiramate (8 patients), Lamotrigine (8 patients), Lacosamide (5 patients), Oxcarbazepine (5 patients), Sulthiame (4 patients), Clonazepam (3 patients), Perampanel (3 patients), Primidione, Carbamazepine, Felbamate and Zonegram (2 patients each), Rufinamide and Ethosuximide (1 patient each). Eleven patients were not treated by any AED during the treatment with CBD.

Patients follow-up duration was 3–33 months (average time-15.6 months; SD 7.23, median time-18 months). Seventeen patients (30%) dropped out from the study; Ten due to bothersome ARs (3 of the 10 patients had more than 80% reduction of seizures), six due to lack of response and one who was unable to take the oil extract.

3.2. Dosing

Oral CBD doses (not including inhalation) ranged from 4 to 32 mg/kg/day (average 11.4 mg/kg/day, SD 5.9). The median CBD calculated dose was 11 mg/kg/day. The total daily dose ranged between 77 and 800 mg (average 305 mg, SD 136). Only 4 patients received more than 20 mg/kg/day and only eleven (19%) patients used more than 400 mg CBD per day.

3.3. Response to treatment

The response rate was evaluated for all seizures types. Of the 46 patients include in the efficacy analysis-Twenty (43.5%) had a seizure reduction of 0–49%, including 2 patients with seizure exacerbation, 10 (22%) had a reduction of 50–75%, 14 (30%) had a reduction of 75–99% and two (4%) were seizure-free. Nineteen patients (41%) partially or completely tapered 1–3 AEDs during the follow-up period due to improvement in seizure rate.

There was no statistical significant difference in terms of response to treatment between patients with ID and those with normal intelligence, different cannabis strain, patients treated with Clobazam vs those not treated with Clobazam, various etiologies or various seizure types, although when excluding seizures types that were recorded in less than four patients each, there was a trend in response rate between various seizure types, which was as followed: atypical absence > primary generalized tonic-clonic seizures > Complex partial seizures > myoclonic seizures > partial seizures with secondary generalization. Patients younger than 10 years at treatment onset had higher improvement rate compared to older patients (78% vs 48%, $p = 0.026$). There was an association between younger age at treatment onset and response to treatment (OR-5.3, $p = 0.013$). We did not demonstrate a statistically significant correlation between improvement rate and CBD calculated dose. However, patients treated with CBD dose of 11 mg/kg/day or more (the median dosage provided) had higher improvement rate compared to patients treated with <11 mg/kg/day (80% vs 50%, $p = 0.043$). Treatment duration in patients with positive response to CBD-enriched oil was longer than for patients without a positive response (20.1 months, SD-5.7 vs. 14.5 months, SD 10.5, $p = 0.01$).

We defined tolerance to CBD-enriched cannabis oil treatment as reduced treatment efficacy following improvement in seizure control in patients who were stable in terms of their AED regimen, and without another identifiable cause for seizure exacerbation. This was observed in 9 patients (16%), 8 of whom during the

Table 1
Demographic and clinical characteristic of the study population (n = 57).

Demographic data	
Age at treatment initiation	9.6 years, SD-4.9 (range 1–20 years)
Patients with Intellectual disability n (%)	43 (77%)
Treatment prior to cannabis oil n (%)	
3–5 AEDs	7 (12%)
6–10 AEDs	30 (53%)
11–17 AEDs	20 (35%)
Ketogenic diet (n)	19 (33%)
Vagal nerve stimulation (n)	18 (31%)
Epilepsy surgery	3 (5%)
Etiology: n (%)	
Cortical malformations	10 (17%)
Early epileptic encephalopathy with known genetic etiology	10 (17%)
Epileptic syndrome	2 (3.5%)
Perinatal insult (Hypoxic-ischemic encephalopathy, prematurity complications, congenital CMV infection)	6 (10%)
Viral encephalitis	5 (9%)
Head trauma	2 (3.5%)
Parenchymal bleeding	1 (1.7%)
Brain Tumor	1 (1.7%)
Unknown etiology	17 (30%)

SD-standard deviation.

first year of treatment. The degree of seizures aggravation was 20–30%. In 4 patients, dose increment regained the former initial positive response, whereas in 2 patients, the additional increment was not possible due to ARs.

3.4. Change in mode of cannabis administration

Seventeen patients were transferred to inhaled cannabis, either alone or in combination with oil extract. Inhaled cannabis was administered via inhalers, vaporizers (Volcano Medic®, PuffiT™), or smoking, dependent on age and level of cooperation of patients. Reasons for incorporating inhalation treatment were the presence of status epilepticus, the presence of a cluster of seizures, inability to take CBD oil orally due to a serious aversion to the taste and oily texture, individual preferences, especially in the adolescents' group, and a desire to increase efficacy. Transfer or addition of inhaled CBD-enriched cannabis caused additional improvement (20–40%) in 6 (35%) of these 17 patients. Most patients reduced the dose of concomitant CBD-enriched oil extract, thus increased efficacy could not be explained solely by the increased total dose of CBD.

3.5. Adverse reactions

Twenty-six (46%) patients had ARs (Table 2). Nine patients had more than one AR. ARs, of which the most common was somnolence, were the main reason for treatment cessation reported in 58% (10/17) of patients who dropped out of the study. Improvement of ARs with time or dose reduction reduced the severity of ARs in the other sixteen patients and enabled treatment continuation. One patient developed Psychosis. This patient was also reported to develop psychosis using other AEDs suggesting a psychiatric predisposition. There was no statistical significant difference between the 2 strains in terms of ARs. All ARs were reversible and resolved within 24 h of treatment withdrawal. A Beneficial response such as improvement in behavior, communication, sleep, and spasticity was reported in 23% of patients.

Table 2
Adverse Reactions.

Adverse reaction n = 57	Frequency	Percent
Somnolence	8	14%
Aggressiveness	5	9%
Loss of appetite (with concomitant weight loss in 2/5)	5	9%
vomiting	5	9%
irritability	4	7%
Weight gain	2	3.5%
Psychosis	1	1.7%
Depression	1	1.7%
Memory loss	1	1.7%
Movement disorder	1	1.7%
Reduced water intake	1	1.7%
Diarrhea	1	1.7%
Seizure exacerbation	2	3.4%

Kaplan–Meier cumulative survival analysis was performed to assess the role of treatment response as well as the presence of adverse reaction on the cumulative probability of CBD- enriched oil treatment duration among patients. This analysis showed that the proportion of participants being treated with CBD in each time point during the study follow up time was higher in patients with positive anti-seizure response to CBD-enriched oil treatment (Breslow-Wilcoxon test for significance, $P < 0.05$). The presence of adverse reactions did not significantly influence the treatment survival time ($p = 0.12$). Prediction of treatment survival by combining various covariates was performed using the Cox regression survival analysis model, including response to treatment and adverse reactions as coefficients (Fig. 1). The treatment survival time was higher for patients with a positive anti-seizure response and without adverse reactions (Exp (B) = 1.883), but this was not statistically significant ($p = 0.174$).

4. Discussion

There is now emerging evidence to suggest the efficacy of cannabis for the treatment of treatment-

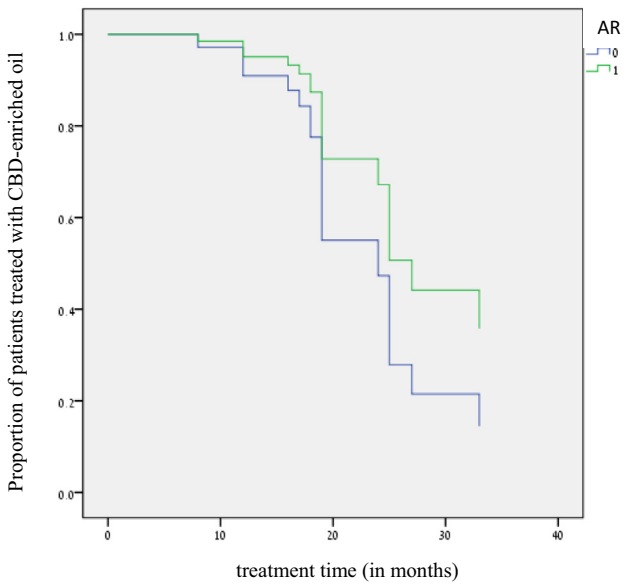

Fig. 1. Cox Proportional Hazard regression survival curves for CBD-enriched medical cannabis treatment. The curves show the proportion of participants being treated with CBD-enriched medical cannabis at each time point during the study follow-up time, in relation to treatment response and adverse reactions. (Exp(B) = 1.883, p = 0.17). AR= adverse reactions. 0 = no reported adverse reactions 1 = at least one reported adverse reaction.

resistance epilepsy in children and young adults. However, the precise mechanism of action is still unclear, and an FDA approval is still lacking for most formulations, forcing physicians to prescribe products produced by marijuana dispensaries, raising questions regarding the uniformity and dosing standardization, which complicate the interpretation of clinical effect. Despite those caveats, there are now attempts to scientifically study cannabis effect and safety profile, as well as standardize its preparations. The current observational study describes our experience in treating children and adolescents with refractory epilepsy treated with high- CBD whole plant extract. Although entry criteria were stringent to include only refractory patients, treatment regimen was flexible and tailored to each patient, therefore, allowing frequent changes of CBD dose and tapering of AEDs. The flexibility of the treatment, despite its methodological limitations, particularly the lack of control group, and self-reporting of the seizure response, allowed for more refined evaluation of significant treatment related concerns, including dosing, response rate, tolerance as well as safety and tolerability, and is less biased than survey studies which are exposed to inherent bias.

4.1. CBD dose

Historic calculated CBD dose for adult patients ranges between 200 and 300 mg/day [4,22]. While initial dose protocols originating from GW Pharmaceutical study (GWP) [24] and different pediatric epilepsy centers suggested increments of up to 25 mg/kg/day, the FDA recently approved daily dose of 50 mg/kg of 99% CBD (Epidiolex, GWP) [24]. The GWP Epidiolex study is one of the largest prospective open-label trial published thus far, using a standardized, purified 99% oil-based CBD extract. A total of 313 patients were recruited and included in the safety and tolerability analysis, of whom 261 who had at least 12 weeks of follow-up were included in the efficacy analysis. CBD Doses were titrated until intolerance or to a maximum dose of 25–50 mg/kg. However, most patients did not reach the maximal dosage of 50 mg/kg/day, and the mean cannabidiol dose at 12 weeks was 22.9 mg/kg in the safety analysis group and 22.7 mg/kg in the efficacy analysis group. In our study, average dose was 11.7 mg/kg/day, and none of the patients' daily dose exceeded 30 mg/kg/day, or a total daily dose of 800 mg.

Cannabinoids are very lipophilic, resulting in decreased bioavailability, which presents challenges in optimal drug delivery. Notably, most of our patients succeeded to keep the oil drops under the tongue for several minutes, thus practicing mucosal absorption, which is three folds higher than oral delivery, and might explain the increased efficacy. We demonstrated higher response rate in patients receiving higher doses of CBD-enriched medical cannabis. This is in concordance with observation published in a recent GW Pharma press release that claimed that 20 mg/kg/d of Epidiolex is more effective than 10 mg/kg/d for the treatment of LGS which was more effective than placebo [28]. However, despite the improved response to higher treatment doses noted in many patients, ARs may prevent some of the patients from maintaining higher dose levels, an observation previously noted by Devinsky et al [24]. Importantly, the amount of THC that can be safely used in pediatric population was never tested. Facing the alarming bells of long-term THC treatment on cognition [12], upper limits for the THC component, especially in children should be considered.

4.2. Seizures reduction

There was a reduction in mean monthly seizure frequency of ≤50% in 56% of the patients, whereas in 35% of patients, reduction rate was ≥75%. This efficacy is comparable to that described by others [24,25]. The efficacy in some patients was higher than recorded in the final report but bothersome ARs necessitate a reduction of CBD-enriched oil dose, thus reducing optional higher efficacy. Similarly, it is plausible that efficacy was also reduced due to the tapering of other AEDs following improvement, due to a potential synergistic effect of other AEDs and cannabis, in particular clobazam, of which blood levels can be increased using CBD as recent

evidence suggest [29,30]. This reported synergistic effect of CBD on clobazam could potentially confound the interpretation of CBD anti-epileptic effect. Unfortunately, we did not perform measurements of Clobazam blood levels during CBD-enriched oil treatment to test this hypothesis. However, there was no difference in terms of response to treatment in patients using Clobazam and those without Clobazam treatment, which could stem from the relatively small cohort. This should be further studied in pharmacodynamics studies, using a strict methodology to measure plasma levels of other AEDs.

We did not demonstrate a difference in terms of treatment efficacy between patients with various syndromes or seizure types. Recent publications address the variable efficacy of cannabis according to different syndromes or seizure types. These include a Facebook questionnaire that showed efficacy among children with Dravet syndrome [23], a questionnaire from Colorado [26], a retrospective study [27], the GWP open – label study by Devinsky et al [24], and a recent open – label study by Hess et al [31] suggesting efficacy in patients with Tuberous sclerosis complex. A Subgroup analysis evaluating improvement per syndrome/etiology in the GWP open – label study [24] revealed maximal improvement in patients with Lennox Gastaut syndrome (71%), followed by tuberous sclerosis (66%), and Dravet syndrome (50%) [28]. Additionally, a Recent phase 3 double – blind placebo-controlled clinical trials studying the efficacy and safety of Epidiolex have shown promising results of reduction of drop attacks in patients with Lennox-Gastaut syndrome and Dravet syndrome, compared to placebo [28,32]. The number of patients with either Dravet syndrome or drop attacks in our study was too small to draw any conclusion, and further data is needed to perform a comprehensive analysis. We are aware that absence, as well as myoclonic seizures, are likely to be the most susceptible to biased reporting and to "placebo effect" counting errors, as these seizures are difficult to identify and quantify. This might explain some of the favorable response seen in our study for this type of seizures.

The development of tolerance following prolonged exposure to cannabis has been previously described in the literature in both animals and human studies and is currently attributed to the adaptive phenomena consisting in pharmacodynamic events (down-regulation/desensitization of cannabinoid receptors) [33–34]. We described tolerance effect in 16% of patients, and were presumably able to minimize this effect by increasing the CBD dose either by increasing oral dose or by switching to inhaled cannabis, which is considered to have an estimated bioavailability of 30% [35], Compared to 6% from oral delivery due to significant first-pass metabolism in the liver. Oral-mucosal/sublingual delivery has bioavailability similar to the oral route but less variability [36]. Inhalation was used as a method for patients with frequent seizures or prolonged daily seizures who are unable to take the drug. This is due to a more rapid onset and plasma delivery (<10 min) compared to oral administration [37]. This administration method is subjected to large inter-subject differences in plasma CBD concentrations due to differences in the depth of inhalation. The preferred route of administration, particularly in the pediatric population, together with tolerance aspects, should be further studied as it affects the pharmacokinetics of CBD.

4.3. Whole plant enriched extract versus single molecule

In the current study, patients were given whole plant mixture with a fixed ratio of CBD/THC. Importantly, the role of the different other cannabinoids and other chemicals of the Marijuana such as terpenoids and flavonoids is not yet clear. The advantage of pharmacological preparations is clear, including quality control, dose standardization, and consistency of the products, enabling to better monitor treatment and ascertaining the safety, efficacy, and drug-drug interactions. However, the complimentary synergism of the different phytocannabinoids as well as yet- unknown other derivatives in the whole plant extract ("entourage effect") mixture is plausible. Some laboratories throughout the world are currently trying to correlate between the preferred mixtures and efficacy for different diseases. It is possible, therefore, that future formulas will contain more than THC and CBD alone. This question should be further studied in the future.

4.4. Adverse reactions

The Rate of ARs in our study was comparable to those of Devinsky et al and Tzadok et al, in which ARs were reported in approximately 46–79% [24,25]. The Most common side effect was Somnolence, reported to be present in 14% of our patients, 23% in the GWP study and 22% in the Israeli cohort. Gastrointestinal ARs were also common in all the above-mentioned studies, potentially related to the presence of cannabinoid receptors throughout the GI tract, which may play a role in the regulation of food intake, nausea and emesis, and GI motility. Remarkably, diarrhea, which might be explained by the apparent partial agonist action of pure cannabidiol on dopamine D2 receptors [38], was less frequently reported in our study compared to the GWP study (1.7% vs 19% respectively). Of the 46 patients, only 2 had seizure exacerbation (including one patient with status epilepticus). This is lower than described in other studies [24]. A possible explanation for this observation could indeed stem from the study methodology and may include a placebo effect, reporting bias etc. As most patients in our study were cognitively impaired,

the cognitive effect of high extract CBD oil could not be fully assessed and remains an enigma, yet to be solved. The beneficial effects reported among patients, including increased alertness, improved communication, sleep quality improvement, and spasticity reduction was described by others [25,26], and is one of the parental motivators for receiving this treatment, oftentimes despite lack of anti-seizure efficacy. Future studies using more objective measures to assess the quality of life would shed light on the various positive effects of cannabis treatment. It is worth noting that due to the methodology of the study, and the fact that our cohort is a continuous registry, with some of the patients enrolled towards the end of the study, the treatment survival curves in Fig. 1 presumably shows higher rates of drop-out whereas it simply reflects follow up time to study termination rather than the actual treatment time.

The major limitation of this study was that it was an uncontrolled open – label rather than placebo-controlled trial. As children tend to have a greater response to placebo compared to adults, with placebo response rate of up to 19% [39], performing placebo-controlled studies is essential, despite its cumbersomeness in this refractory population. This is of great importance particularly in the face of the increased parental interest in cannabis for epilepsy and the high level of expectation regarding the therapeutic potential of cannabis for the treatment of epilepsy. Other limitations include self-reporting of the seizures, non-monitoring of concomitant AEDs levels and other laboratory data such as liver function tests, the inability to measure CBD dose and correlate with treatment response, and the lack of electroencephalographic data. However, we believe that the data presented here reflects an ongoing observation of relatively prolonged follow-up time, which enabled describing CBD treatment-related dilemmas over time in a clinical setting, including adverse reactions and tolerance. This, despite methodology-related caveats, could be of benefit for the clinician who is treating epilepsy with cannabis.

To conclude, CBD-enriched cannabis extract is shaping up to be a very promising anti-seizure option, showing excellent results compared with other AEDs for refractory patients. In concordance with recent encouraging evidence, this open – label study using parental report, showed that CBD- enriched cannabis extract appears to have potential anti-seizure effect as an add-on treatment in pediatric patients with refractory epilepsy, with favorable safety profile. We strongly support the need for controlled studies of cannabis products using consistent formulations and strict methodology to assess the effectiveness of various compounds in the plant, to analyze pharmacodynamics as well as pharmacokinetic aspects, and to test the long-term efficacy of cannabis, given our preliminary results of a tolerance effect. We advocate the need for assessing the cognitive effect of cannabis on the developing brain, by performing formal neurocognitive assessments. The Neuroprotective and Anti-Inflammatory mechanism by which CBD potentially controls seizure activity should be further explored.

Author contribution

Moran Hausman Kedem: contributed to the interpretation and analysis of the results, wrote the manuscript.

Shay Menascu: cared for the patients.

Uri Kramer: conceived and planned the study, cared for the patients, contributed to data interpretation and analysis, edited the manuscript.

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